

Multiscale Simulations of OLED Molecules: Quantum Optimizations, Comparative Analysis, and Optical Gap Estimations Across Time

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Abstract

Our research investigated the relationship between molecular structure and OLED performance using multiscale simulations. Employing B3LYP/6-31G(d,p) quantum calculations, we analyzed electronic properties and optical gaps of DOBNA, CzPA, and 3phCN-ICz molecules. Molecular dynamics simulations, conducted with the OPLSaa force field, explored structural impacts on color purity over time. Integration of data through Python analysis revealed minimal molecular interaction effects on emitted color based on our approximations. By comparing computational simulations with experimental data, our study provided valuable insights into the design and optimization of OLED molecules.

Introduction

Organic molecules have become focal points in energy research due to their distinctive properties and wide-ranging applications. Derived from carbon-based compounds, these molecules offer unique characteristics that render them highly sought-after for various energy-related pursuits.

Organic Photovoltaics (OPVs) represent promising avenues for renewable energy generation, utilizing sunlight to produce electricity. These devices function through a series of steps, starting with photon absorption, leading to exciton generation within organic semiconductor materials. Excitons migrate through the material, facilitating charge separation at the donor-acceptor interface, and ultimately contributing to electrical power generation. OPVs offer advantages such as flexibility, lightweight, and cost-effectiveness compared to conventional silicon-based solar cells (Sudheendran Swayamprabha et al., 2020).

Organic Light-Emitting Diodes (OLEDs) leverage organic materials to emit light efficiently, providing energy-efficient lighting and display solutions with vibrant colors and potential flexibility. OLEDs operate by utilizing an anode and cathode connected to an organic semiconducting material. The injected charges recombine within the material, forming excitons that undergo radiative decay, emitting photons (Zhang et al., 2016). The emitted photon's wavelength determines the light's color. While the fundamental gap (LUMO - HOMO) is not the sole determinant of the wavelength of the emitted photon, it can be used as an approximation.

In our research, we propose that the structural design of organic molecules significantly influences OLED device efficiency, particularly in terms of luminescence, optical gap, and color purity. To investigate this hypothesis, we employ multiscale computational simulations, integrating quantum calculations and molecular dynamics.

Quantum calculations enable us to estimate HOMO and LUMO values for the provided molecules using the B3LYP hybrid functional and a 6-31G(d,p) basis set. These values allow us to estimate the optical gap, providing insights into the molecules' electronic properties and their potential for light emission.

Molecular dynamics simulations enable exploration of how molecular interactions influence color purity in OLED devices. By modeling molecular behavior over time, we simulate trajectories and observe structural changes' impact on light emission. This approach offers insights into molecular performance using classical physics principles.

We integrate data obtained from specific time points in molecular dynamics simulations into our quantum calculations to determine the fundamental gap's temporal variations. By analyzing this data across time using Python, we gain deeper insights into how structural changes affect frequency and light emission.

Our study aims to provide valuable insights into OLED molecule design and optimization by comparing computational simulations with experimental data. To advance future computational research in smart energy technologies, we will enhance our skills in quantum chemistry, molecular dynamics simulations, programming (Python), data analysis, and interdisciplinary collaboration.

Methods

The structures of 3phCN-ICz and DOBNA were drawn in the Gaussian 16 software suite, while CzPA was taken from the ligpargen database and converted into a Gaussian Input File (.gjf file). Quantum geometry optimizations and frequency calculations were conducted on each molecule with Gaussian using the B3LYP hybrid functional in conjunction with the 6-31G(d,p) basis set, a widely accepted method for predicting electronic properties of organic molecules. We verified there were no imaginary frequencies for each molecule, confirming that none of the geometries were stuck in a saddle point. We collected the data on the HOMO and LUMO values of the optimized geometries and calculated their fundamental gaps and compared our results to experimental results.

Then, molecular dynamics simulations were conducted in a 5x5x5 nm box with Periodic Boundaries, simulated under room temperature (298 K) for both NVT and NPT, and 1 atm pressure for NPT. Each simulation ran for 8,000 ps utilizing the Optimized Potentials for Liquid Simulations - all atom (OPLSaa) force field. The simulations were completed up to the NVT ensemble, then the NPT ensemble. Each molecule was placed parallel to a duplicate of itself at a distance of approximately 0.4 nm to ensure interactions between the molecules. The simulations

were visualized using Visual Molecular Dynamics 1.9.3. We extracted data on the kinetic energies, potential energies, total energies, and temperatures of each molecule across time during the NPT simulations in the form of .xvg files. We plotted this data using Python 3.12.2 along with libraries pandas 2.2.2, matplotlib.pyplot 3.8.0, and numpy 1.26.4.

Then, for each molecule, 5 frames were taken from the NPT MD simulations, with each frame separated by 2 nanoseconds. For each frame, we conducted single-point quantum optimizations (again using B3LYP/6-31G(d,p)) to determine the HOMO and LUMO values, as well as the fundamental gap values, at these points in time. We plotted these values across time using Python and the same libraries to see the effects of interactions on the fundamental gaps. This gave us an estimation on how molecular interactions affected the color purity.

The research methodologies employed in our study aimed to ensure reproducibility and rigor in our experiments. For quantum optimization, we utilized Gaussview 6.1.1 and Slurm Workload Manager 23.02.5 for job submission. The calculations were performed using the B3LYP hybrid functional in conjunction with the 6-31G(d,p) basis set, a widely accepted method for predicting electronic properties (DFT level of theory). For molecular dynamics simulations, we utilized GROMACS 2022.5 and Visual Molecular Dynamics 1.9.3.

The molecular dynamics simulations were conducted in a 5x5x5 nm box with Periodic Boundaries, simulated under room temperature (298 K) for both NVT and NPT, and 1 atm pressure for NPT. Each simulation ran for 8,000 ps utilizing the Optimized Potentials for Liquid Simulations - all atom (OPLSaa) force field. The simulations were completed up to the NPT ensemble, with molecules placed parallel to each other at a distance of approximately 0.4 nm. The simulations were visualized using Visual Molecular Dynamics.

For data analysis and visualization, Python 3.12.2 was utilized, along with libraries pandas 2.2.2, matplotlib.pyplot 3.8.0, and numpy 1.26.4. The results were tabulated and graphed using Python, including the extraction of molecular structures from every two nanoseconds during the molecular dynamics NPT simulation and single-point quantum calculations to determine HOMO and LUMO values. These methods ensured comprehensive exploration and analysis of the electronic and structural properties of the studied molecules, facilitating meaningful comparisons with experimental data. The Python and Excel files can be found in the Supporting Information section.

Results and Discussion

In this section, we determined the HOMO, LUMO, and Fundamental Gap values for 3phCN-ICz, CzPA, and DOBNA. In Figure 1, the HOMO area is situated on the right side of the molecules, indicating the location of the donor section within the molecule. Conversely, the LUMO area is found on the left side of the molecule, signifying the location of the acceptor section. This visualization aligns with the anticipated distribution of electron density, associating electron-rich regions with the HOMO and electron-poor regions with the LUMO, consistent with molecular orbital theory.

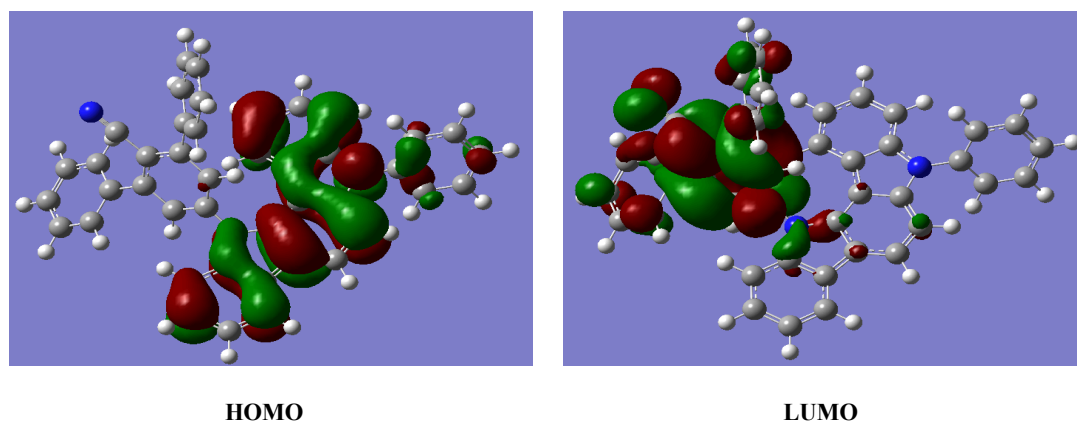


Fig. 1. HOMO and LUMO visual representations of 3phCN-ICz. The colored sections represent the electron density distribution within specific molecular orbitals. Red areas indicate higher electron density. Conversely, green areas indicate lower electron density.

Our results are presented in Table 1.A, while the simulation outcomes conducted by Dyes and Pigments are displayed in Table 1.B. The HOMO values exhibit remarkable similarity, differing by merely 0.34 eV. However, a more substantial variance is observed in the LUMO values, with a notable difference of 0.57 eV compared to those reported by Dyes and Pigments. Notably, our fundamental values are nearly identical, demonstrating a disparity of only 0.22 eV.

HOMO	-5.18 eV
LUMO	-1.63 eV
FUNDAMENTAL GAP	3.54 eV

A

HOMO	-5.52 eV	B
LUMO	-2.2 eV	
FUNDAMENTAL GAP	3.32 eV	

Table 1. Table 1.A represents our quantum calculations results. Table 1.B represents Dyes and Pigments results (Jiang et al., 2023)

For the CzPA molecules, both the HOMO and LUMO regions are predominantly localized in the middle-right section of the molecule, with minimal presence in the left side. This spatial distribution contrasts with the previous molecules studied, where the HOMO and LUMO regions were predominantly situated on opposite ends. The observed localization of both HOMO and LUMO sections in the middle suggests a unique electronic structure for CzPA, possibly influenced by the arrangement of functional groups or molecular symmetry.

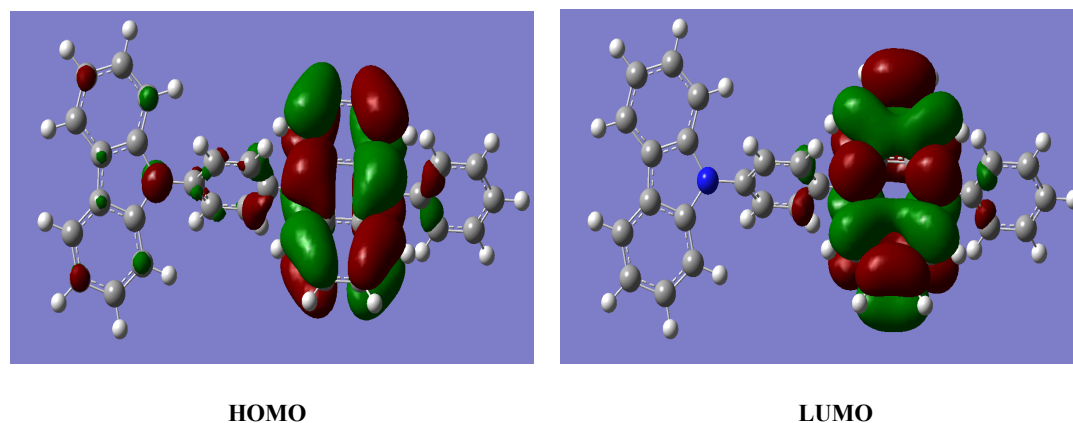


Fig. 2. HOMO and LUMO visual representations of CzPA. The colored sections represent the electron density distribution within specific molecular orbitals. Red areas indicate higher electron density. Conversely, green areas indicate lower electron density.

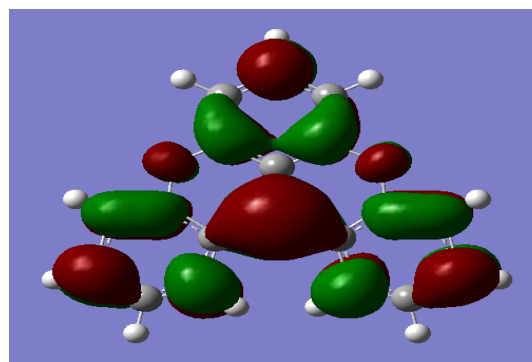
Similar to Table 1, Table 2.A presents our research findings, while Table 2.B displays results reported by Nature Communications. A moderate discrepancy of 0.26 eV is observed in the HOMO values between the two datasets. In contrast, a notably higher difference of 0.35 eV is evident in the LUMO values. Interestingly, the fundamental gap demonstrates minimal disparity, with an inconsequential 0.11 eV difference.

HOMO	-5.54 eV	A
LUMO	-1.90 eV	
FUNDAMENTAL GAP	3.64 eV	

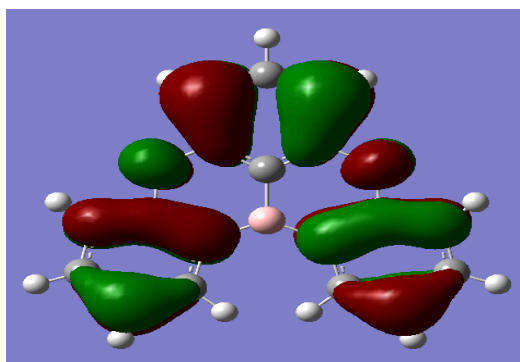
HOMO	-5.80 eV	B
LUMO	-2.25 eV	
FUNDAMENTAL GAP	3.55 eV	

Table 2. Table 2.A represents our quantum calculations results. Table 2.B represents Nature Communications results (Salehi et al., 2019)

Lastly, the observation that both the HOMO and LUMO distributions in the DOBNA molecule are nearly uniform across the molecule, with the exception of the HOMO covering the central section, likely arises from the unique electronic structure conferred by the presence of boron in that region. Boron, being an electron-deficient element, tends to exhibit different electronic properties compared to the surrounding carbon and nitrogen atoms. As a result, the HOMO, which represents electron-rich regions, may be localized around the boron atom, while the LUMO, indicating electron-deficient regions, is distributed more evenly across the molecule.



HOMO



LUMO

Fig. 3. HOMO and LUMO visual representations of DOBNA. The colored sections represent the electron density distribution within specific molecular orbitals. Red areas indicate higher electron density. Conversely, green areas indicate lower electron density.

Table 3.A presents our research findings, while Table 3.B displays results from Organic Electronic. Notably, DOBNA exhibits the most substantial deviation in difference values among the three molecules studied. Despite the minimal disparity of 0.03 eV in HOMO values between the tables, a stark contrast is evident in the LUMO values, with a notable difference of 0.94 eV. This discrepancy in LUMO values may arise from differences in computational methodologies, such as the treatment of molecular interactions or the choice of basis set. Consequently, the observed 0.87 eV difference in Fundamental Gap values underscores the critical importance of methodological consistency and meticulous validation when comparing computational results across studies.

HOMO	-5.64 eV	A
LUMO	-1.68 eV	
FUNDAMENTAL GAP	3.96 eV	

HOMO	-5.61 eV	B
LUMO	-2.62 eV	
FUNDAMENTAL GAP	3.09 eV	

Table 3. Table 3.A represents our quantum calculations results. Table 3.B represents Organic Electronics results (Ha et al., 2023)

Figure 4.A illustrates the parallel alignment of 3phCN-ICz molecules at an approximate distance of 0.4 nm. Throughout the NVT simulation, as depicted in Figure 4.B, there is observable divergence in the molecular positions, suggesting a progressive separation between adjacent molecules. A similar trend is evident in the NPT simulation, as depicted in Figure 4.C. This divergence in molecular positions indicates a repulsive interaction between neighboring molecules. Notably, Figure 4.C depicts molecules extending beyond the simulation box;

however, this occurrence is inconsequential due to the utilization of a simulation box with periodic boundary conditions, ensuring an accurate representation of molecular interactions within the system.

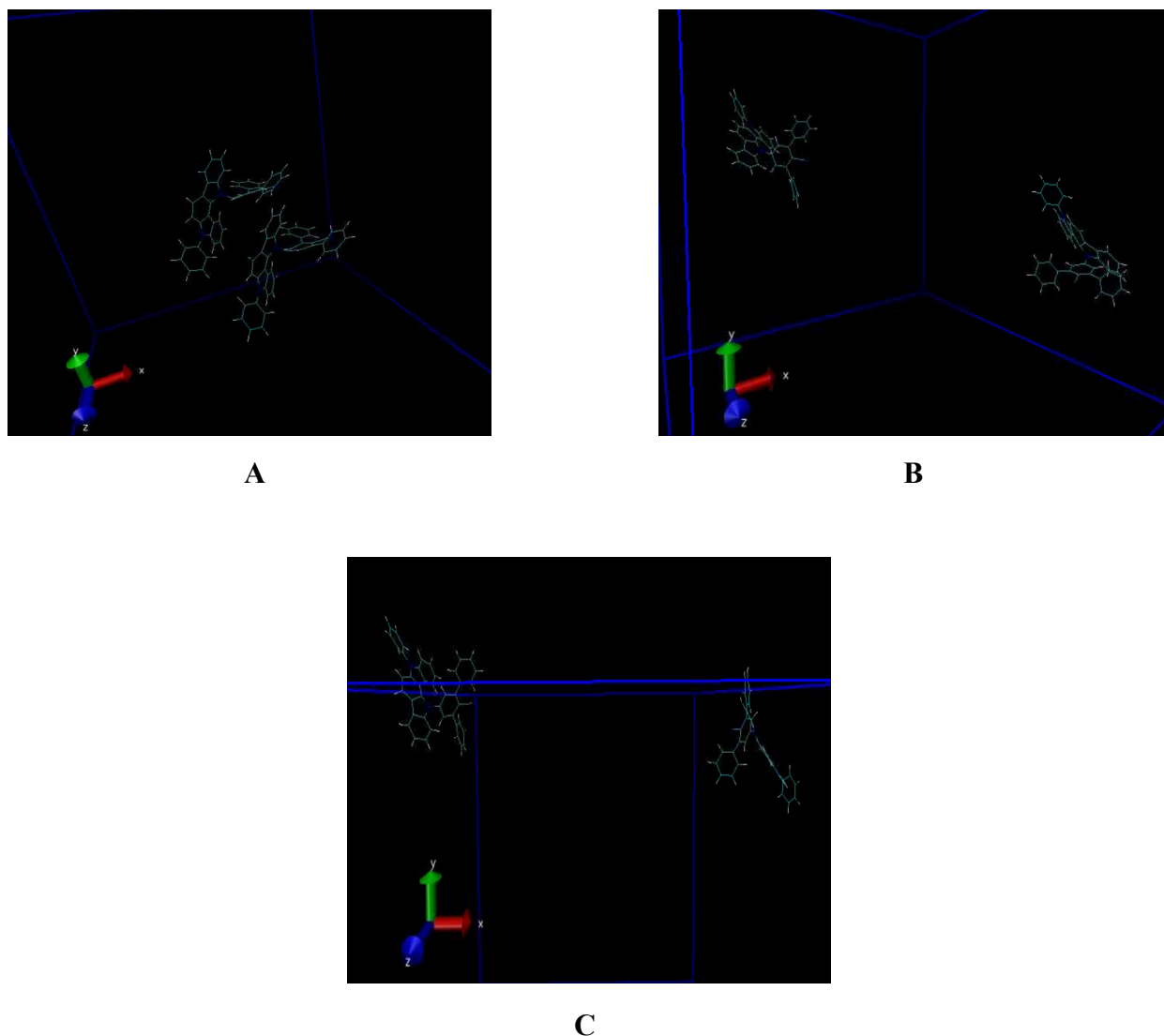
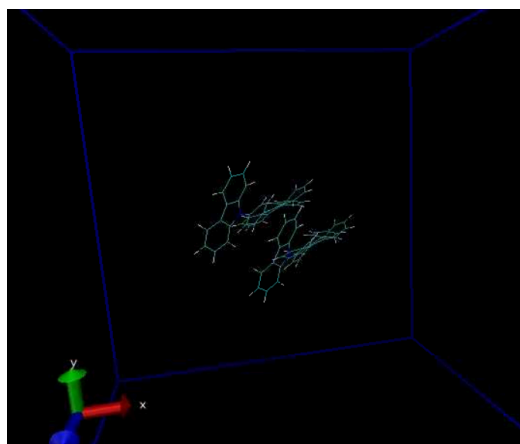


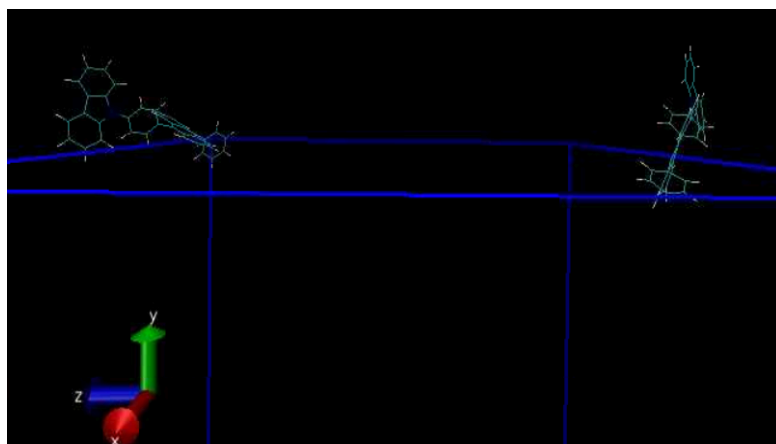
Fig. 4. Figure 4.A represents the molecule 3phCN-ICz before interactions. Figure 4.B represents 3phCN-ICz after NVT. Figure 4.C represents 3phCN-ICz after NPT.

Figure 5.A depicts the parallel arrangement of CzPA molecules, situated at an approximate distance of 0.4 nm. Notably, during the NVT simulation, as depicted in Figure 4.B, the molecules exhibit a trend of converging towards each other. This convergence is attributed to the periodic boundary conditions utilized in the simulation, whereby molecules positioned on opposite sides

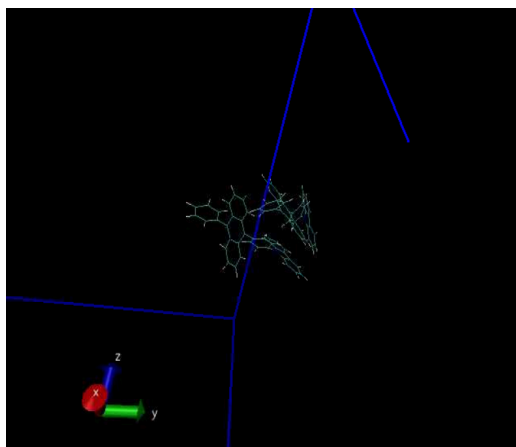
of the simulation box effectively interact. Consistently, a similar pattern is observed in the NPT simulation, as evidenced in Figure 4.C. These observations collectively suggest an attractive interaction between neighboring molecules, leading to their propensity to approach one another within the simulated system.



A



B

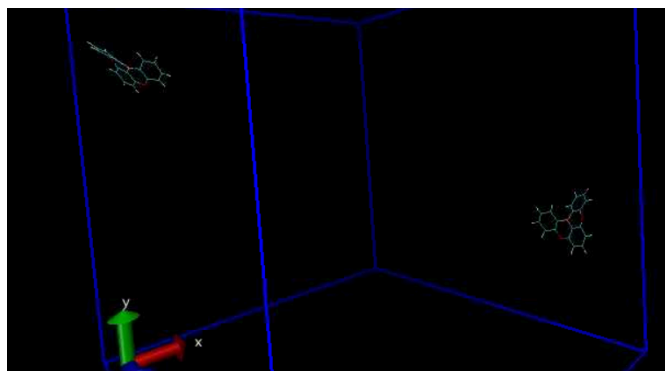


C

Fig. 5. Figure 5.A represents the molecule CzPA before interactions. Figure 5.B represents CzPA after NVT. Figure 5.C represents CzPA after NPT.

In Figure 6.A, the parallel arrangement of DOBNA molecules is presented, situated at an approximate distance of 0.4 nm. Notably, during the NVT simulation, as illustrated in Figure 4.B, these molecules exhibit a propensity to draw closer to each other. This behavior is attributed

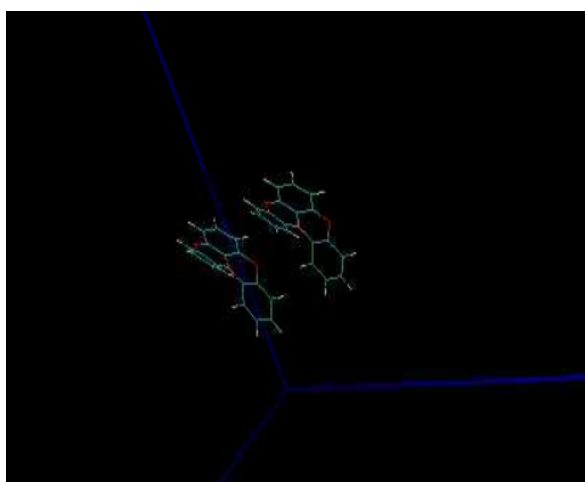
to their positioning on opposite sides of the simulation box within the context of periodic boundary conditions. Consistently, a similar pattern of convergence is observed in the NPT simulation, as evidenced in Figure 4.C. These observations collectively support the inference of an attractive interaction between the DOBNA molecules, prompting them to approach each other within the simulated system.



B



C



A

Fig. 6. Figure 6.A represents the molecule DOBNA before interactions. Figure 6.B represents DOBNA after NVT. Figure 6.C represents DOBNA after NPT.

After conducting molecular dynamics simulations, we extracted energy and temperature data for the three molecules over the course of 8000 ps. In Figure 7.A, the energy profile of the 3phCN-ICz molecule is depicted. Total energy fluctuates between 2000 and 1800 kJ/mol, with potential energy varying from 1500 to 1600 kJ/mol and kinetic energy ranging from 300 to 400 kJ/mol. Notably, all three energy components exhibit a relatively constant trend, indicated by

nearly flat lines. Figure 7.B illustrates the temperature dynamics over time, with temperature points scattered across the plot. However, the majority of temperatures fall within the range of 350 to 250 Kelvin.

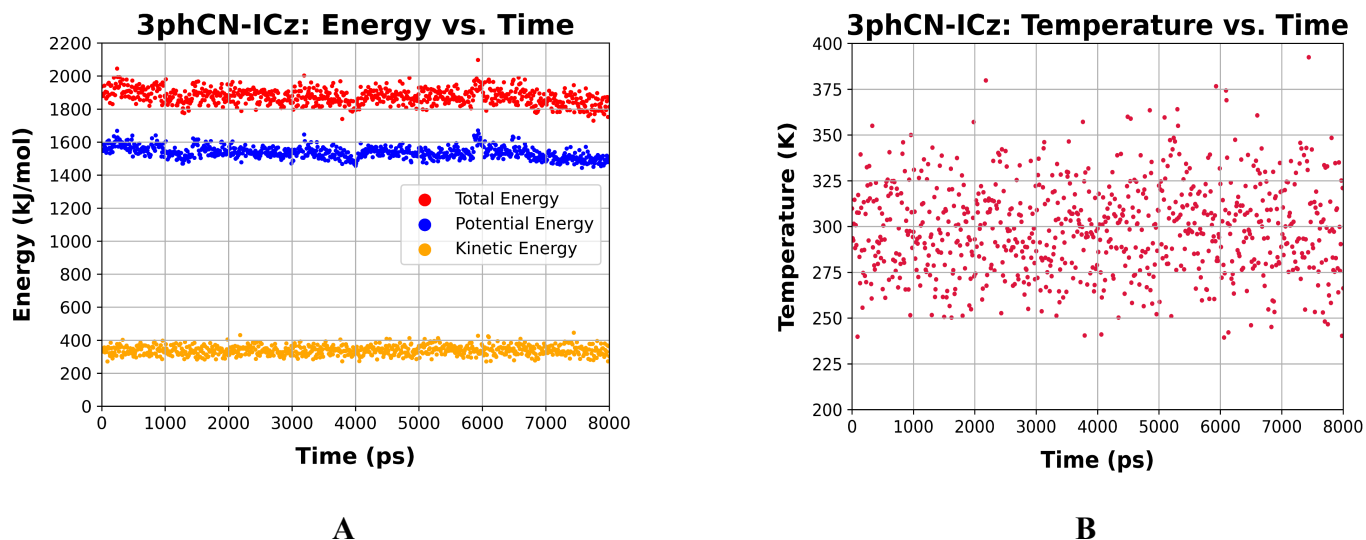
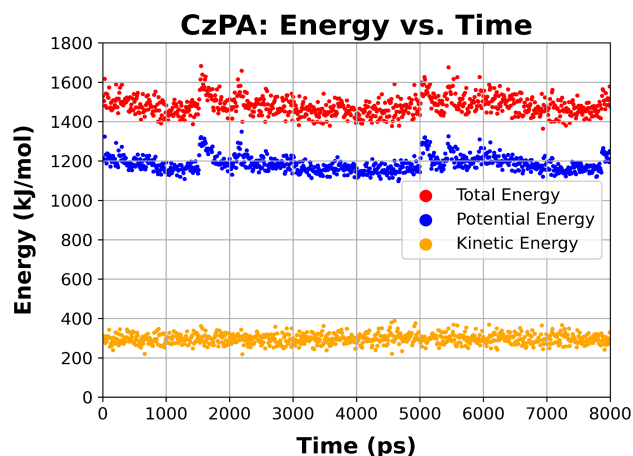
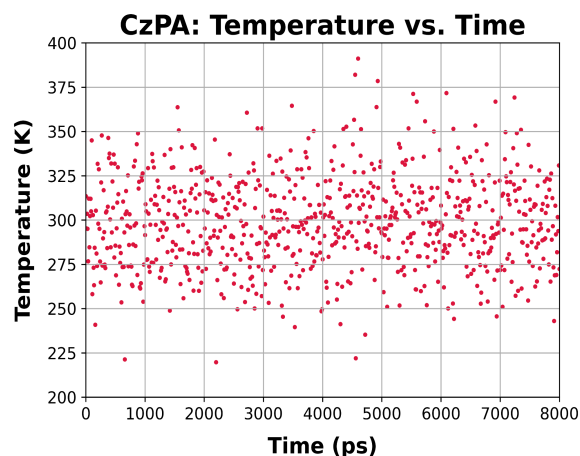


Fig. 7. Figure 7.A represents the energy (Total, Potential and Kinetic) of 3phCN-ICz across time in kJ/mol. Figure 7.B represents the temperature in kelvin of 3phCN-ICz across time.

Similarly, Table 8.A presents the total, potential, and kinetic energies of CzPA. Total energy values range between 900 and 800 kJ/mol, with the majority of the trend maintaining a flat line. However, small peaks are observed around 1500-2000 ps and between 5000 and 6000 ps. These peaks may indicate transient fluctuations in energy, possibly due to molecular rearrangements or interactions. Likewise, potential energy values range between 1200 and 1300 kJ/mol, exhibiting similar small peaks during the aforementioned time frames. Meanwhile, kinetic energy remains relatively constant, ranging from 300 to 350 kJ/mol. In Table 8.B, temperature values are scattered across time, with the majority falling between 250 and 350 Kelvin.



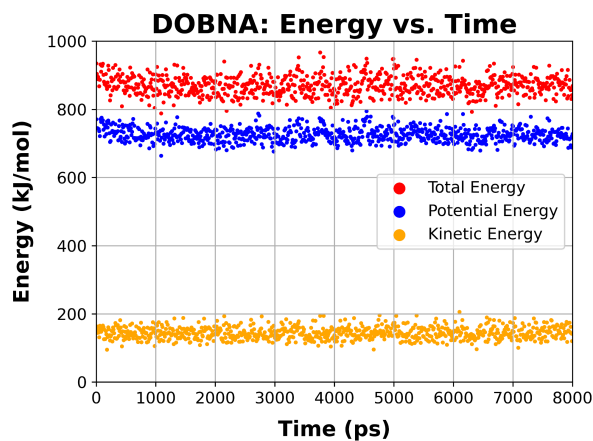
A



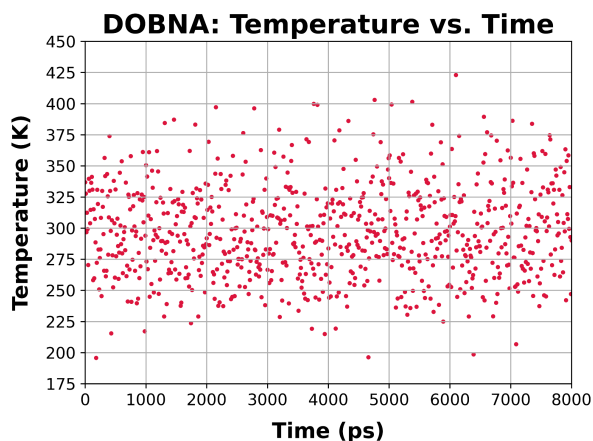
B

Fig. 8. Figure 8.A represents the energy (Total, Potential and Kinetic) of CzPA across time in kJ/mol. Figure 8.B represents the temperature in kelvin of CzPA across time.

Lastly, Figure 9.A illustrates the total, potential, and kinetic energies of DOBNA over time. Notably, all three energy components exhibit a consistent, flat trend. Total energy values range from 800 to 900 kJ/mol, with potential energy between 700 and 750 kJ/mol, and kinetic energy ranging from 100 to 200 kJ/mol. The steady nature of these energy profiles suggests a stable molecular system with minimal energy fluctuations throughout the simulation period. In Figure 9.B, temperature values for the molecule are depicted over time, with the majority falling within the range of 225 to 400 Kelvin.



A



B

Fig. 9. Figure 9.A represents the energy (Total, Potential and Kinetic) of DOBNA across time in kJ/mol. Figure 9.B represents the temperature in kelvin of DOBNA across time.

After analyzing the total energies of the three molecules, we identified a significant trend regarding their molecular sizes. As depicted in Figure 10, the molecule with the highest total energy corresponds to the largest molecule, 3phCN-ICz, with energy values ranging between 1800 and 2000 kJ/mol. Conversely, the medium-sized molecule, CzPA, exhibits total energy values ranging between 1400 and 1600 kJ/mol. Lastly, the smallest molecule, DOBNA, demonstrates the lowest total energy, ranging between 800 and 900 kJ/mol. This trend highlights the influence of molecular size on the overall energy profile of the molecules

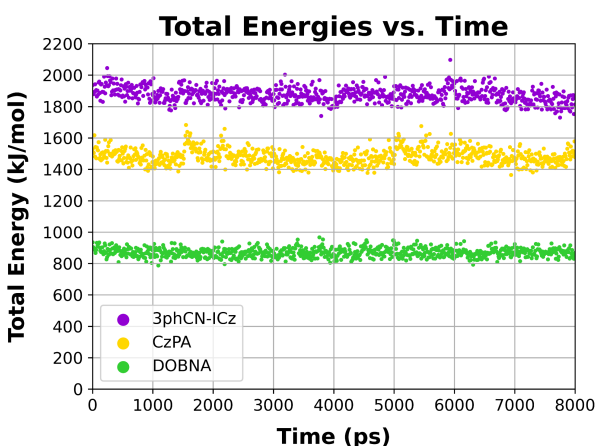


Fig. 10. Plot representation of the total energies of 3phCN-ICz, CzPA and DOBNA across time.

After conducting molecular dynamics simulations and analyzing the energy vs. time and temperature vs. time profiles, we extracted the HOMO, LUMO, and Fundamental Gap values across five frames of time, 0 ns, 2 ns, 4 ns, 6 ns and 8 ns. For 3phCN-ICz in Table 4, the HOMO values remained constant at -5.14 eV throughout the simulation period. Similarly, the LUMO values exhibited minimal variation, with values of -1.51 eV for the first three time frames and -1.47 eV for the last two frames, resulting in a negligible difference of 0.04 eV. Consequently, the Fundamental Gap values remained relatively stable, with a value of 3.63 eV for the first three time frames and 3.67 eV for the last two frames.

3phCN-ICz	HOMO (eV)	LUMO (eV)	Fundamental Gap (eV)
0 ns	-5.14	-1.51	3.63
2 ns	-5.14	-1.51	3.63
4 ns	-5.14	-1.51	3.63
6 ns	-5.14	-1.47	3.67
8 ns	-5.14	-1.47	3.67

Table 4. Tabulated HOMO, LUMO and Fundamental Gap Values across 5 frames in time of 3phCN-ICz

As depicted in Figure 11, the HOMO values exhibit consistency across all time frames, remaining constant throughout the simulation period. In contrast, both the LUMO and Fundamental Gap values demonstrate a slight increase from the 6 ns onward.

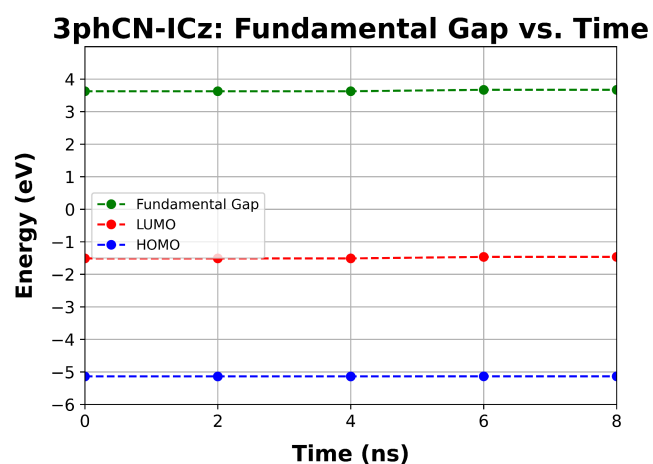


Fig. 11. Plot representation of HOMO, LUMO, and Fundamental Gap values of 3phCN-ICz across 5 frames of time

For the CzPA molecule, in Table 5 the HOMO values demonstrate relative stability, with values of -5.021 eV for the first three time frames and -4.977 eV for the last two, resulting in a minimal difference of 0.04 eV. While the LUMO values remain constant for the first three time frames at -2.065 eV, they show a slight decrease to -2.072 eV for the last two time frames. Consequently, the Fundamental Gap values also exhibit nearly constant behavior, with values of 2.956 eV for the first three time frames and 2.905 eV for the last two.

CzPA	HOMO (eV)	LUMO (eV)	Fundamental Gap (eV)
0 ns	-5.021	-2.065	2.956
2 ns	-5.021	-2.065	2.956
4 ns	-5.021	-2.065	2.956
6 ns	-4.977	-2.072	2.905
8 ns	-4.977	-2.072	2.905

Table 5. Tabulated HOMO, LUMO and Fundamental Gap Values across 5 frames in time of CzPA

As depicted in Figure 12, the LUMO values exhibit stability for the first 3 time frames, but it decreases after 6ns. Additionally, both the Fundamental Gap and HOMO values demonstrate a decrease and an increase respectively after 6 ns.

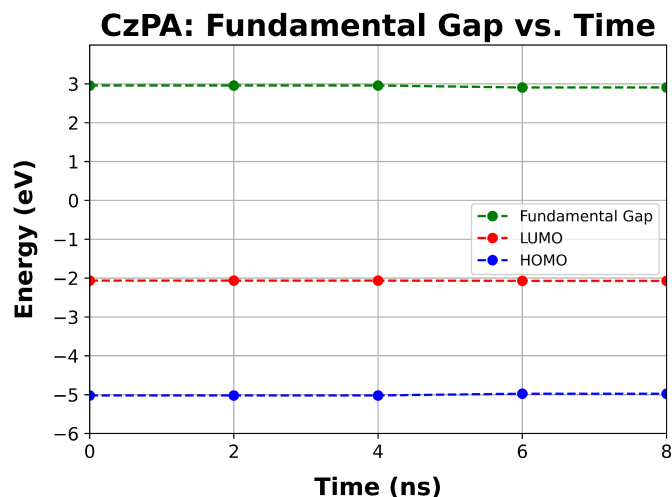


Fig. 12. Plot representation of HOMO, LUMO, and Fundamental Gap values of CzPA across 5 frames of time

Table 6 illustrates the electronic properties of DOBNA over time. The HOMO values demonstrate consistency, with values of -5.82 eV for the first three time frames and -5.89 eV for the last two, resulting in a minimal difference of 0.03 eV. The LUMO values remain constant throughout the simulation with a value of -1.50 eV. Interestingly, the Fundamental Gap values display a slight decrease, with values of 4.32 eV for the first three time frames and 4.30 eV for the last two.

DOBNA	HOMO (eV)	LUMO (eV)	Fundamental Gap (eV)
0 ns	-5.82	-1.50	4.32
2 ns	-5.82	-1.50	4.32
4 ns	-5.82	-1.50	4.32
6 ns	-5.79	-1.50	4.30
8 ns	-5.79	-1.50	4.30

Table 6. Tabulated HOMO, LUMO and Fundamental Gap Values across 5 frames in time of DOBNA

In Figure 13, the HOMO and Fundamental Gap values exhibit an increase and decrease respectively at 6 ns, while the LUMO values remain constant throughout the simulation period.

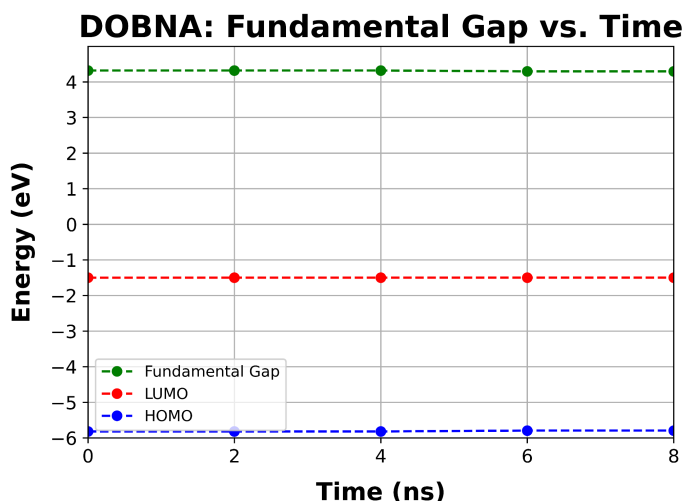


Fig. 13. Plot representation of HOMO, LUMO, and Fundamental Gap values of DOBNA across 5 frames of time

In Figure 14, we depicted the fundamental gaps of the three molecules over time. Initially, all molecules maintain constant values until the sixth ns. Subsequently, slight variations are observed: 3phCN-ICz exhibits a slight increase, while CzPA demonstrates a slight decrease. In contrast, DOBNA, though not significantly, exhibits a slightly more pronounced decrease compared to CzPA. This observation prompts further inquiry into the underlying factors driving these changes. Regarding the influence of molecular size, while it's generally accepted that molecular size can affect electronic properties, the observed changes suggest that other factors, such as molecular interactions or structural dynamics, may also play a significant role.

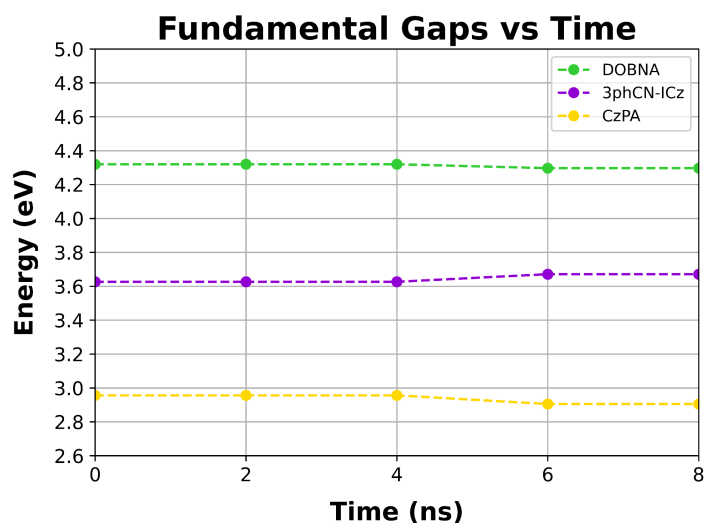


Fig. 14. Plot representation of the Fundamental Gap values of 3phCN-ICz, CzPA and DOBNA across 5 frames of time

Conclusions

In conclusion, our research aimed to explore the impact of structural design on the efficiency of OLED devices, focusing on molecular dynamics and quantum calculations through multiscale computational simulations, we have gathered valuable insights into the electronic properties and behaviors of three organic molecules: 3phCN-ICz, CzPA, and DOBNA.

Our findings revealed several significant observations:

1. **Electronic Properties:** We determined the HOMO, LUMO, and Fundamental Gap values for each molecule, noting both similarities and disparities compared to existing literature. We determined that our approximations of the fundamental gaps of our molecules align closely with experimental results in the cases of 3phCN-ICz and CzPA, but not with DOBNA.
2. **Molecular Dynamics Behaviors:** Through molecular dynamics simulations, we observed distinct behaviors for each molecule. While 3phCN-ICz exhibited a trend of divergence between adjacent molecules, indicating repulsive interactions, both CzPA and DOBNA demonstrated a tendency to converge, suggesting attractive interactions within the simulated system.
3. **Influence of Molecular Size:** Analysis of total energy profiles revealed a correlation between molecular size and energy, with larger molecules exhibiting higher total energy values. This highlights the significance of molecular size in determining overall energy profiles and potentially influencing device efficiency.

Furthermore, our investigation led us to the realization that our initial hypothesis regarding the profound influence of molecular structural changes due to molecular interactions on color purity was unfounded. Contrary to our expectations, our estimations indicate that molecular interactions have minimal impact on color purity.

However, while our research contributes valuable insights, further investigations are warranted to address certain limitations and areas for improvement:

1. **Methodological Consistency:** Disparities between our results and existing literature suggest the need for methodological consistency and meticulous validation in future studies. Standardizing computational methodologies and benchmarking against experimental data could enhance the reliability and reproducibility of findings.
2. **Molecular Interactions:** Our simulations revealed diverse molecular dynamics behaviors, indicating the complex interplay of intermolecular interactions. Further

research focusing on understanding and manipulating these interactions could lead to enhanced control over device properties and performance.

In summary, while our research provides greater insight in understanding the relationship between molecular structure and OLED device efficiency, continued efforts are necessary to address remaining challenges and refine our understanding of organic materials for optoelectronic applications.

Supporting Information

 Supporting Information

<https://drive.google.com/drive/folders/1ouKMWNgrVX1-rwVLKDJfH8dn3OGosqhN?usp=sharing>

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